CALCIUM CARBONATE

Take with meals

Separate from many drugs

CHEMICAL NAME $CaCO_3$ (40% elemental calcium: 1.25 g = 500 mg calcium)
CLASS Calcium supplement; antacid; phosphate binder
TRADE NAMES Calcichew · Caltrate · Tums · Adcal-D3 (with vitamin D)

THERAPEUTIC USES / INDICATIONS

- Calcium deficiency; osteoporosis (with vitamin D and bone therapy).
- Hyperphosphataemia in CKD (phosphate binder).
- Antacid for heartburn/dyspepsia.
- Hypocalcaemia (mild, chronic); pregnancy (pre-eclampsia prevention in low-calcium diets).

ADMINISTRATION

- **PO** chewable tablets or effervescent; with meals (acid aids absorption; required for phosphate binding).
- Divide doses ≤500 mg elemental calcium for better absorption.
- Separate from interacting drugs by 2–4 h.

DOSAGES

- **Osteoporosis:** 1–1.2 g elemental calcium/day (diet + supplement), usually 500–1,000 mg supplement with vitamin D 800–1,000 IU.
- **Phosphate binder:** 1–2 g (as carbonate) with each meal, adjusted to phosphate.
- **Antacid:** 500–1,000 mg as needed (max 7 g $CaCO_3$/day).

PHARMACOKINETICS

- Absorbed in small intestine (~25–35%; vitamin D-dependent; ↓ with achlorhydria/PPIs — calcium citrate better).
- Excreted in faeces and urine.

PHARMACODYNAMICS

- Provides **calcium** for bone mineralisation, nerve and muscle function.
- Binds dietary **phosphate** in gut → insoluble calcium phosphate.
- Neutralises gastric acid.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Constipation, bloating, flatulence.
- Belching (CO_2 release).
- Nausea.

ADVERSE EFFECTS (SERIOUS)

- Hypercalcaemia, milk–alkali syndrome (with high doses + alkali).
- Kidney stones.
- Possible ↑ cardiovascular events (high-dose supplements — debated).
- Vascular calcification in CKD (excess).

CONTRAINDICATIONS & CAUTIONS

- Hypercalcaemia, hypercalciuria.
- Calcium kidney stones (relative).
- Severe renal impairment (caution, monitoring).

KEY INTERACTIONS

- ↓ absorption of levothyroxine, tetracyclines, fluoroquinolones, iron, bisphosphonates, zinc; separate by 2–4 h.
- **Thiazides:** ↑ calcium.
- **Digoxin:** arrhythmias with hypercalcaemia.

NURSING MANAGEMENT

- Monitor serum calcium, phosphate (CKD), vitamin D.
- Schedule doses with meals and away from interacting drugs.
- Manage constipation.

PATIENT EDUCATION

- Chew tablets and take with meals.
- Keep thyroid tablets, iron, and some antibiotics at least 2–4 hours apart.
- Increase fluids and fibre.
- Get calcium from diet too (dairy, leafy greens).

KEY POINTS

- 40% elemental calcium.
- With meals; separate from many drugs.
- Osteoporosis, CKD phosphate binder, antacid.
- Constipation; hypercalcaemia.